

DU Mathsoc Problem Solving: Set 5

Date: 13/10/2026

~Beginners Paper~

For any assistance or inquiries about the problem sets, please email me at: quizmaster.maths@csc.tcd.ie.

Problem 1. Prove that $\{x \in \mathbb{N} : x = 7a + 3, a \in \mathbb{N}\} \cap \{y \in \mathbb{N} : y = 6b + 4, b \in \mathbb{N}\} = \{z \in \mathbb{N} : z = 42c + 10, c \in \mathbb{N}\}$.

Problem 2. Determine the real numbers x, y, z that satisfy $x^2 - yz = y^2 - zx = z^2 - xy$ and $xy + yz + xz = 2025$.

Problem 3. Solve the following system of equations over the real numbers:

$$\begin{cases} \lfloor \sqrt{\lfloor x \rfloor} \rfloor = \sqrt{\lfloor \sqrt{y} \rfloor} \\ \lfloor \sqrt{\lfloor y \rfloor} \rfloor = \sqrt{\lfloor \sqrt{z} \rfloor} \\ \lfloor \sqrt{\lfloor z \rfloor} \rfloor = \sqrt{\lfloor \sqrt{x} \rfloor} \end{cases}.$$

Problem 4. Let $a, b, c > 0$. Prove that $\frac{a^2}{b^2} + \frac{b^2}{c^2} + \frac{c^2}{a^2} \geq \frac{a}{c} + \frac{b}{a} + \frac{c}{b}$.

Problem 5. Solve the equation $\frac{5}{x - \lfloor x \rfloor} = \frac{6}{x} + \frac{4}{\lfloor x \rfloor}$.